RSM — Just-Physics Chain (v9.2)

Rendering register throughout. Every claim is a correspondence, :: , meaning maps-to and never is; no entry carries evidential weight in either direction. A physics chain under this discipline is a map plus a refusals ledger. Every entry states a break-condition — what would kill the mapping; an entry without one does not enter. Tags: [flagship] strongest instance; [P2] correspondence; [held light] suggestive, unpressed; [prediction] falsifiable — the predictions section is the chain's only evidential section.

The banned sentence class, named at the door: no identification of the frame boundary with a physical event. The branching is not an event; time is frame-internal; a date belongs inside the frame it would explain. The permitted maximum: *read from inside by cosmology, the frame boundary renders as a beginning.*

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1. The constants cluster

1_n :: $\hbar/2$ *[flagship]*. The complementarity floor ($\sigma_a \cdot \sigma_b \geq 1_n/2$) and the uncertainty relation ($\Delta x \cdot \Delta p \geq \hbar/2$) share inequality-shape and units: the conserved product of conjugate spreads carries the dimensions of action. The frame's unit renders as the quantum of action. The shared saturation structure is not a third ground: the framework's floor is saturated by the minimal traversal, which lives in the signed chart and is [rendering], so on that axis the correspondence runs between a physical result and a result about the framework's chart. Mechanism note, load-bearing: quantum mechanics derives its floor from Fourier duality and noncommutativity; the framework from floor + symmetry + winding — no transform, no noncommutativity. Same shape, different engines; the mapping is between results, not derivations, and the difference is the finding. **Break:** a physical conjugate pair with a product floor untraceable to any exclusion structure; or the winding-measure premise shown underivable.

The slice invariant :: c *[flagship]*. The split action $e^{j\phi}$ is the algebra of Lorentz boosts — SO(1,1), hyperbolic rotation preserving Q_j — and the boost is a rescaling of the conjugate pair at fixed invariant (scale as motion along G, in the laboratory). **The null cone $Q_j = 0$ :: the light cone**; the unreachable terminals :: no massive body reaches c — "c is a boundary, not a speed," with an algebraic backbone. **Break:** any demonstration that the boost group's structural seat requires more than the two-slice identity supplies. Named strain: the slice gives signature (1,1); physics has (1,3) — inherited by the dimension item.

2. The seat statement: $E = mc^2$

E and m are not a conjugate pair (that reading is refused — one conserved quantity in two unit systems is not a trade). The typing that holds: **m :: the frame unit** (1_n :: $(mc^2)^2$), **(E, pc) :: the split coordinates**, and **$E = mc^2$ :: the Amplitude Floor** — $E \geq mc^2$, equality exactly at rest: the seat. The Amplitude Floor is the boundary of the mode-expressible domain, not a limitation on it, and $E \geq mc^2$ is a floor of exactly that kind. It cascades: **m = 0 is Q_j = 0 — massless :: the null cone.** Photons live on the cone because a zero-unit frame has no seat and no rest; the null case is the degenerate limit, not an excluded terminal. The rest-state ray $E/m = c^2$ :: the lineage-axis — one line threading the seats of all particle-frames, root-capped. **Break:** a massive system saturating below its floor; or a massless system exhibiting a rest frame.

3. Wick and the two slices [flagship]

L = T − V :: Q_j (the difference form; the gradient slice); H = T + V :: Q_i (the sum form; the circle slice); Wick rotation :: the slice exchange — e^{iS/ℏ} ↔ e^{−βH}. Physics treats the rotation as a computational device; the rendering reads it as the two-slice identity wearing spacetime signature — and the circle slice is a rendering on the framework's side too, so the correspondence runs between two things of the same type. The variational principle enters modally, not teleologically: the stationary path is not chosen; it is where contributions fail to cancel. **Break:** obstructions to the continuation in curved or interacting settings that the two-slice identity cannot carry.

4. Complementarity; spin

One law, two exclusive readings, coinciding only at the seat :: Bohr complementarity — with the addition Copenhagen declined: what the two descriptions are readings *of*. *[P2]*

The two returns :: the spinor double cover *[held light]*: invariant at π, position at 2π (the framework) ↔ sign at 2π, state at 4π (spin-½). The framework side is [rendering], so this is a correspondence between a physical structure and a chart feature. **The debt, stated plainly:** the 2π phase of spin-½ is measured (neutron interferometry). The sign is orientation-of-reading, and that measurement is owed an account. **Break:** the two returns shown to be parametrization artifacts.

Candidate account, not entered. The parturition map's two preimages are one branch compared from the child's side of the joint — an indexed comparison performed across a frame boundary. On this reading a measured sign is evidence not of a second obtainer but of a comparison made from a register the measurement is inside. It has no break-condition, and by the register header an entry without one does not enter. Recorded here as the shape an account would have to take.

5. Excluded cores

Circulation organized around an unoccupied core: the free vortex ($v \cdot r$ = const — the gradient in flow, with the constitutive guard: nothing was removed from the core; the exclusion is not historical), orbital structure, horizon interiors *[held light — the interior as the chart's display of a center the frame cannot express; anything stronger is the banned class]*. **The s-orbital strain, kept visible:** quantum mechanics places the electron's density maximum *at the nucleus* for every s-state — the center, in physics' own best description, is not spatially avoided there. Either O's exclusion is not positional in the atomic rendering (phase- or measure-register instead), or this correspondence class is weaker than it looks. First-order strain against the framework's most important object; the exposure is the entry.

6. Frames and the lineage :: effective theories and the renormalization group [P2, strong]

A frame :: an effective theory; 1_n :: the theory's own cutoff-relative unit; the lineage parameter c :: the RG scale μ; index-relationality :: scheme-dependence. Physics discovered frame-by-frame description on its own: each scale its own degrees of freedom, no final theory required downward. Directionality flag: RG flow runs child→parent (coarse-graining); generation runs parent→child; the maps are inverse and must be stated as such. Named strains: RG flow is continuous, frames constitutively discrete; UV fixed points are terminals, which the framework denies — if a physical fixed point is a genuine end of the lineage, the rendering breaks there. **The Planck floor, stated as the disagreement it is:** if reality has a minimum length, P1's antecedent is false *as physics*, and every within-physics rendering loses its referent while the conditional stands (a conditional is not refuted by its antecedent failing). The closest thing the framework has to a falsifier-shaped exposure.

7. Irreversibility

Generation is irreversible (reversal would require P_0 to obtain) :: the thermodynamic arrow. Mechanism note as the entry: the physical arrow is statistical, the structural one modal; probability and impossibility are different modalities; the mapping is between directions, not reasons. **Open exposure — the two arrows:** parturition-irreversibility versus frame-scale shear are rival renderings of the arrow, unresolved; the break-test is within-frame entropy increase with no generation event, which thermodynamic relaxation appears to supply. The oldest live physics question in the project.

8. The Bekenstein sort [open exposure — the chain's honest wound]

A naive cross-frame accounting gives capacity scaling with volume; holography says area. If the cross-frame unit relation is ever closed, its first test is whether it produces area scaling — and if it produces volume, the rendering program takes real damage. The threat is recorded at full strength precisely because it is the sharpest adversarial fact known against the program. The constitutive share between parent and child is pattern and reference, with no transfer; whether any **quantitative** relation accompanies it is open. If the answer is eventually "none," this exposure converts from threat to silence — the program would produce neither area nor volume scaling because it would have no cross-frame accounting at all. That is a coherent outcome and a weaker one, and it would be recorded as going silent rather than surviving.

9. Quasiperiodicity

If the cross-frame ratio is irrational, lineage traversal winds incommensurately — return without repetition :: quasicrystalline order, orbital resonance structure. *[held light; the unit-relation's one observable face, and conditional on that relation existing at all]*

Predictions [the only evidential section]

1. **Unitarity as the floor's dynamical face.** The framework's floor forbids distinction-destruction; objective-collapse theories (GRW-class) predict its violation. A confirmed objective collapse breaks the rendering — a genuine kill-condition.
2. **Structural recession** *[unbuilt]*: the distinguishing observational signature of frame-generation against standard cosmology's dark-energy account — named as owed, not claimed.
3. **Power-law nesting** *[unbuilt]*: scale-distribution structure in nested systems; operationalization owed.

Refusals and deferrals (standing)

Deferred: gravity as recursion-geometry. No route from the skeleton to sourced curvature currently exists, and nothing enters without one. This is a statement about the present state of the work, not about the framework's reach; it is re-openable by building a route, and a warped-frame geodesic construction is the natural place a route would start. **No entry:** consciousness (nothing in the skeleton selects experience). **Retired:** charge conjugation — carried while the signed register was open, and the signed register is chart. **Refused**

typing: $E = mc^2$ as conjugate trade (§2 carries the correct typing). **Parked:** dark matter, dark energy. **Excluded class:** additive conservation laws as renderings of G — a sum admits terminals; listing energy budgets under the gradient would erase the joint's own class-selection argument. **Banned:** the Big Bang identification, named at the door; the maximum stands — read from inside, the frame boundary renders as a beginning, and the manifold-excision reading (the singularity as excised point: unmet origin, not early event) is its strongest general-relativistic face.

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v9.2 — the chains state the framework as currently known; the editorial record holds how it got here.